In [ ]:
```
# EXPERIMENT - 6
## Data Visualization with Seaborn
PIN No.: 241UDS0111

### Aim
To visualize and analyze a dataset using Seaborn plots such as scatter plots, KDE plots, histograms, count plots,
bar plots, boxplots, strip plots, joint plots, and heatmaps.

### Prerequisites
1. Python installed
2. Jupyter Notebook
3. Knowledge of Pandas, NumPy, Matplotlib, and Seaborn
```

In [ ]:
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## Step 1 – Import the Required Libraries & Dataset
```

In [1]:
```
# Import the required libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

# Load the dataset (keep the CSV in the same folder as this notebook)
df_241UDS0111 = pd.read_csv("Experiment_6_241UDS0111_Public_Library_Visits.csv")

# Display the dataset
df_241UDS0111.head()
```

Out[1]:

	city	date	visits	state
0	Hyderabad	2022-01-05	42377	Telangana
1	Chennai	2022-01-05	40722	Tamil Nadu
2	Mumbai	2022-01-05	43399	Maharashtra
3	Bengaluru	2022-01-05	37754	Karnataka
4	New Delhi	2022-01-05	40138	Delhi

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## Step 2 – Info of the Dataset
```

In [ ]:
```
# Display dataset information
df_241UDS0111.info()
```

In [ ]:
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## Step 3 – State-wise Data Points
```

In [2]:
```
# Count the number of records for each state
df_241UDS0111.value_counts("state")
```

Out[2]:
```
state
Telangana      16
Tamil Nadu     16
Maharashtra    16
Karnataka      16
Delhi          16
Name: count, dtype: int64
```

In [ ]:
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## Step 4 – Convert Date Column to Datetime
```

In [3]:
```
# Convert the date column to datetime format
df_241UDS0111["date"] = pd.to_datetime(df_241UDS0111["date"])

# Create a year column for heatmap analysis
df_241UDS0111["year_241UDS0111"] = df_241UDS0111["date"].dt.year

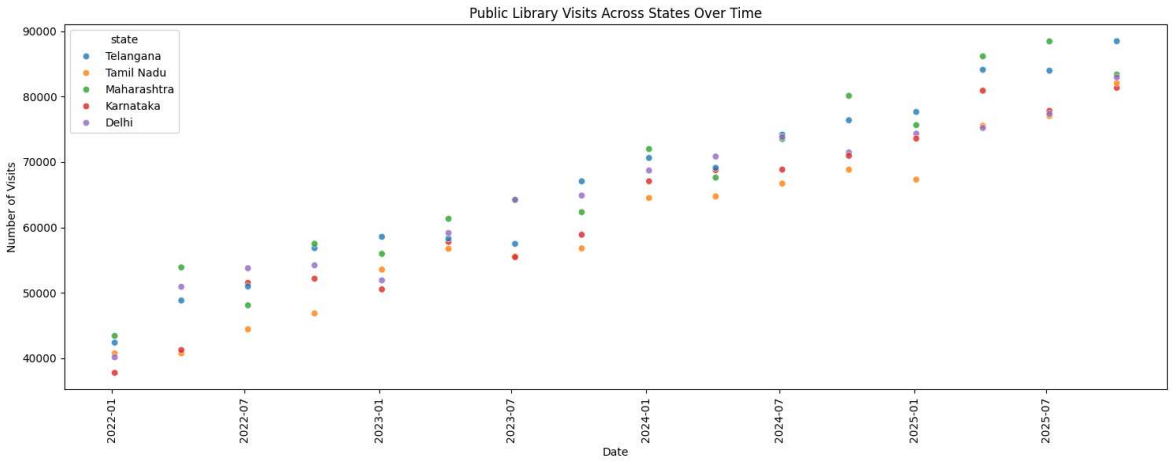
df_241UDS0111.head()
```

Out[3]:

	city	date	visits	state	year_241UDS0111
0	Hyderabad	2022-01-05	42377	Telangana	2022
1	Chennai	2022-01-05	40722	Tamil Nadu	2022
2	Mumbai	2022-01-05	43399	Maharashtra	2022
3	Bengaluru	2022-01-05	37754	Karnataka	2022
4	New Delhi	2022-01-05	40138	Delhi	2022

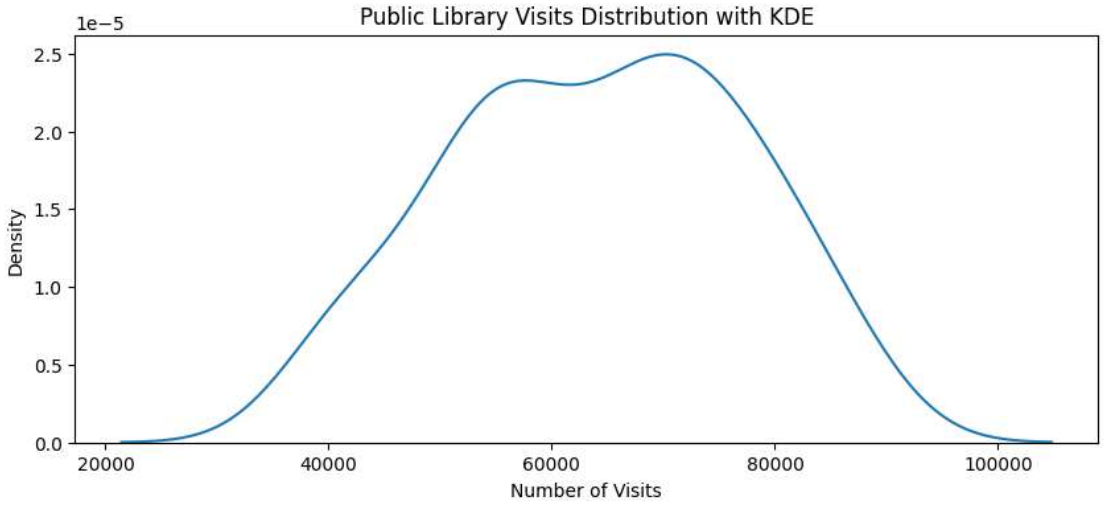
In [ ]:
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## Step 5 – Visualize Data Using Various Plots: Scatter Plot
```

In [4]:
```
# Scatter plot: library visits across states over time
plt.figure(figsize=(15, 6))
sns.scatterplot(
    x="date",
    y="visits",
    hue="state",
    data=df_241UDS0111,
    alpha=0.8
)
plt.title("Public Library Visits Across States Over Time")
plt.xticks(rotation=90)
plt.xlabel("Date")
plt.ylabel("Number of Visits")
plt.tight_layout()
plt.show()
```



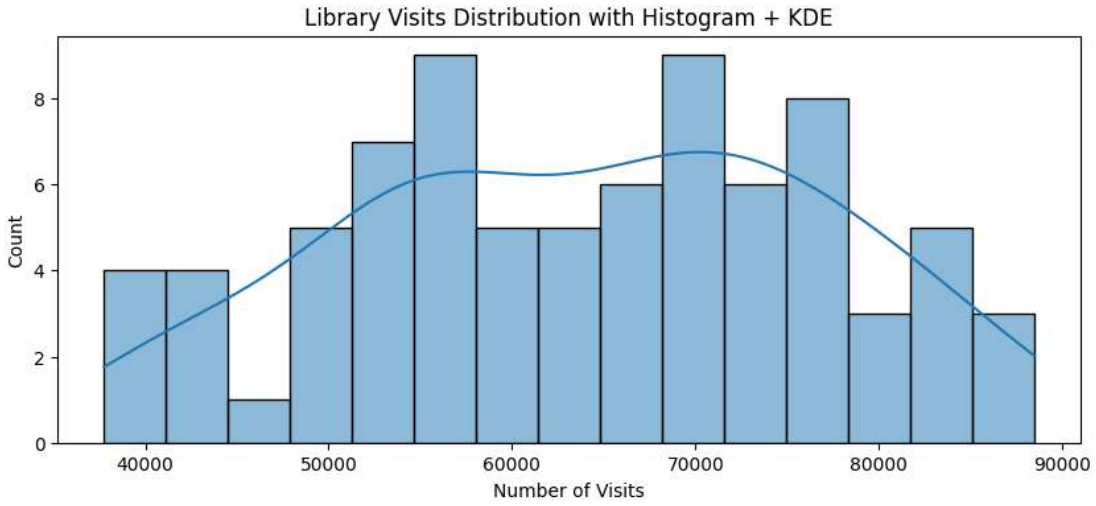
In [ ]:
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## Step 6 – Histograms & KDE: KDE Plot
```

In [5]:
```
# KDE plot: distribution of library visits
plt.figure(figsize=(10, 4))
sns.kdeplot(data=df_241UDS0111, x="visits")
plt.title("Public Library Visits Distribution with KDE")
plt.xlabel("Number of Visits")
plt.ylabel("Density")
plt.show()
```



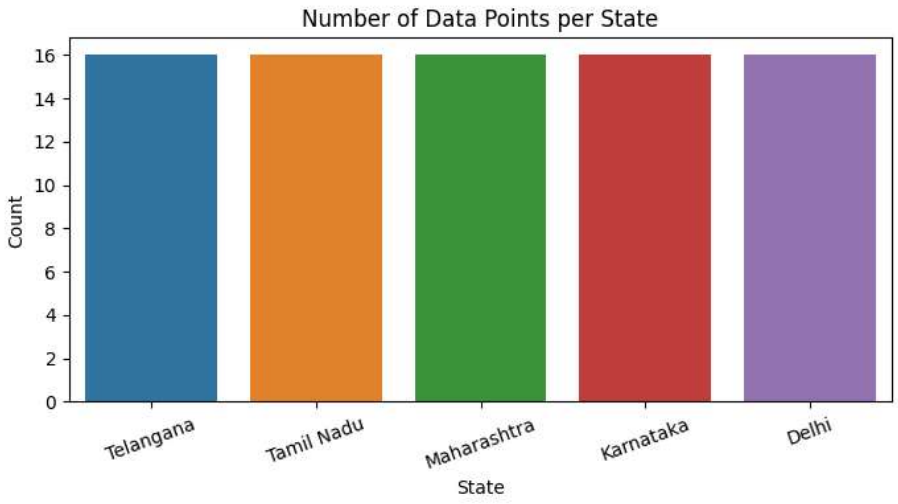
In [ ]:
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## Step 7 — Histogram + KDE
```

In [6]:
```
# Histogram with a KDE curve
plt.figure(figsize=(10, 4))
sns.histplot(data=df_241UDS0111, x="visits", bins=15, kde=True)
plt.title("Library Visits Distribution with Histogram + KDE")
plt.xlabel("Number of Visits")
plt.ylabel("Count")
plt.show()
```



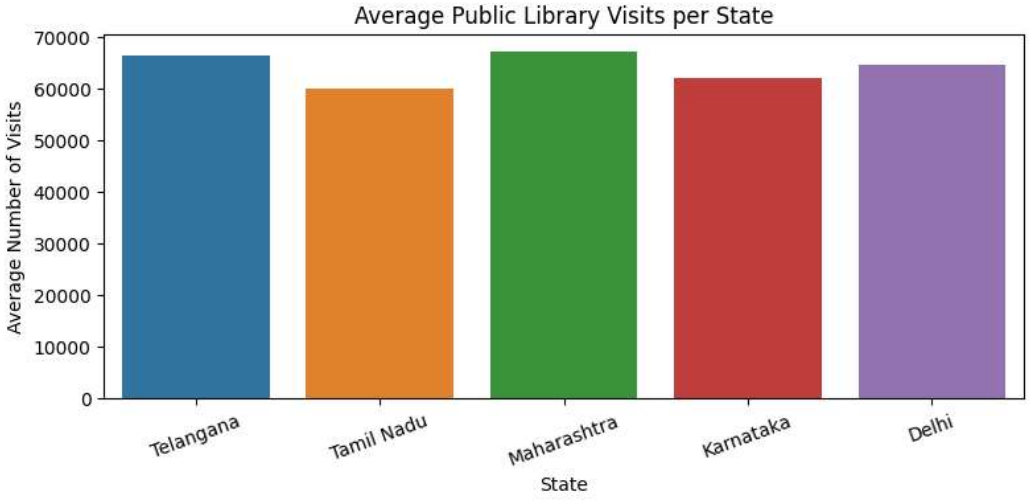
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## Step 8 — Count Plot
```

In [7]:
```
# Count plot: number of records for each state
plt.figure(figsize=(7, 4))
sns.countplot(data=df_241UDS0111, x="state", hue="state", legend=False)
plt.title("Number of Data Points per State")
plt.xlabel("State")
plt.ylabel("Count")
plt.xticks(rotation=20)
plt.tight_layout()
plt.show()
```



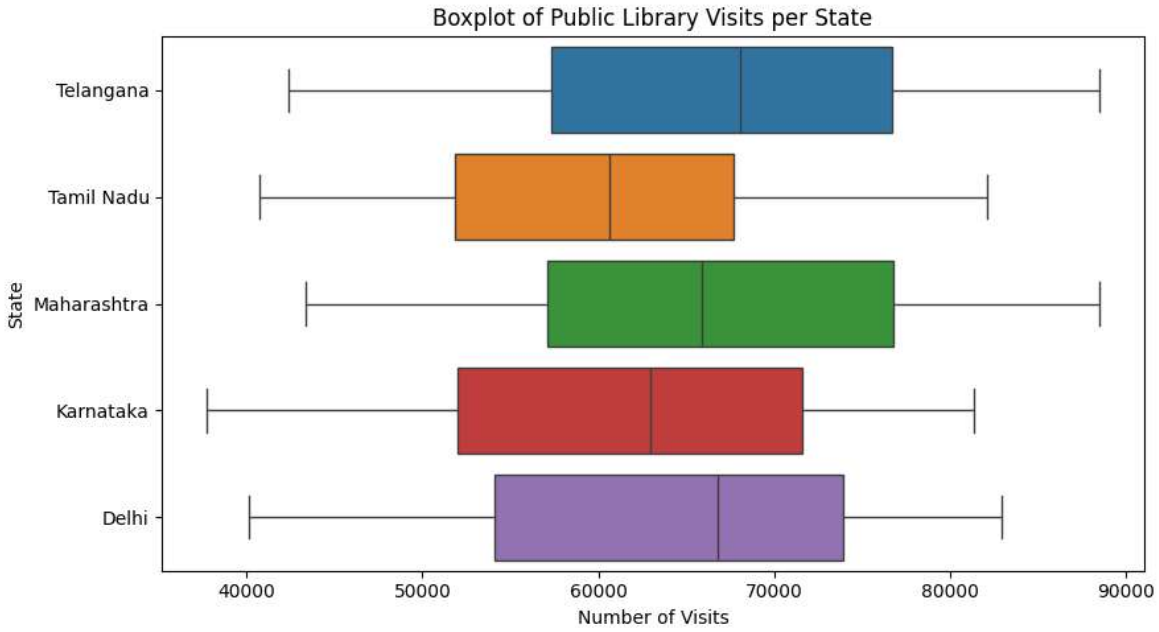
In [ ]:
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## Step 9 — Bar Plot
```

In [8]:
```
# Bar plot: average library visits per state
plt.figure(figsize=(8, 4))
sns.barplot(
    x="state",
    y="visits",
    data=df_241UDS0111,
    hue="state",
    estimator=np.mean,
    errorbar=None,
    legend=False
)
plt.title("Average Public Library Visits per State")
plt.xlabel("State")
plt.ylabel("Average Number of Visits")
plt.xticks(rotation=20)
plt.tight_layout()
plt.show()
```



In [ ]:
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## Step 10 — Boxplot
```

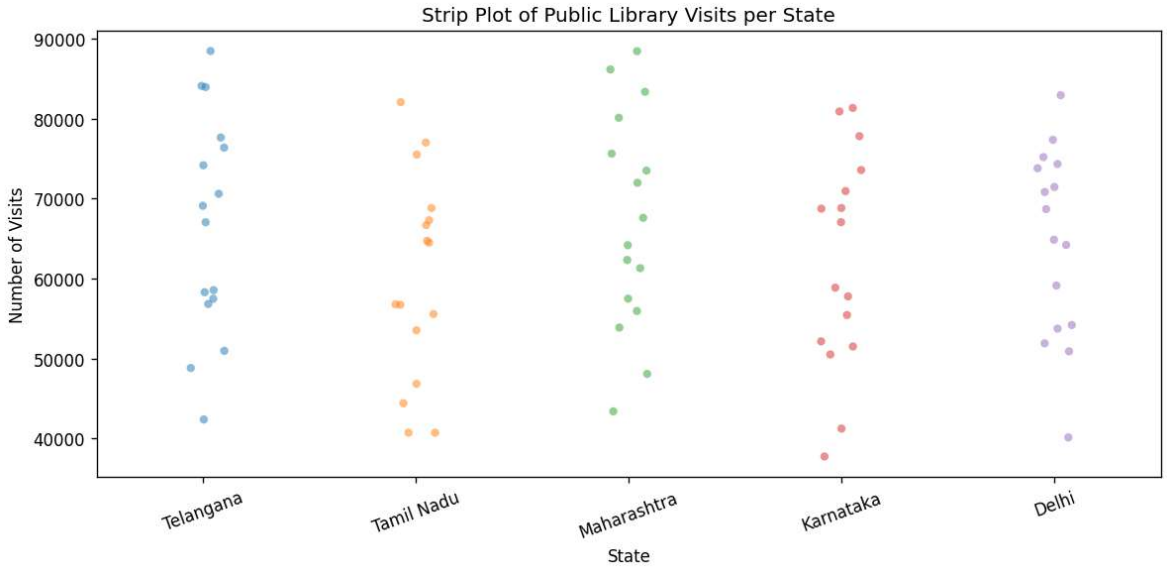
In [9]:
```
# Boxplot: compare distributions and identify possible outliers
plt.figure(figsize=(9, 5), dpi=100)
sns.boxplot(
    x="visits",
    y="state",
    data=df_241UDS0111,
    hue="state",
    legend=False
)
plt.title("Boxplot of Public Library Visits per State")
plt.xlabel("Number of Visits")
plt.ylabel("State")
plt.tight_layout()
plt.show()
```



In [ ]:
```
## Step 11 — Swarm / Strip Plot
```

In [10]:

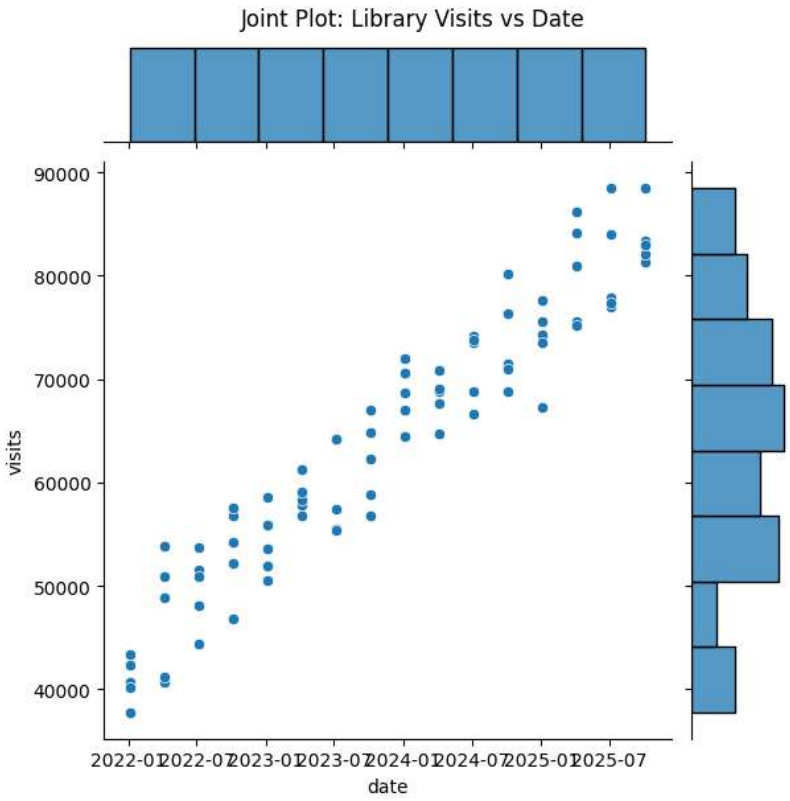
```
# Strip plot: show individual visit-count observations by state
plt.figure(figsize=(10, 5), dpi=120)
sns.stripplot(
    x="state",
    y="visits",
    data=df_241UDS0111,
    hue="state",
    alpha=0.5,
    legend=False
)
plt.title("Strip Plot of Public Library Visits per State")
plt.xlabel("State")
plt.ylabel("Number of Visits")
plt.xticks(rotation=20)
plt.tight_layout()
plt.show()
```



In [ ]:

```
## Step 12 — Joint Plot
```

In [11]:
```
# Joint plot: relationship between date and number of visits
sns.jointplot(
    x="date",
    y="visits",
    data=df_241UDS0111,
    kind="scatter"
)
plt.suptitle("Joint Plot: Library Visits vs Date", y=1.02)
plt.show()
```



In [ ]:
```
## Step 13 — Heat Map: Average Visits per Year per State
```

In [12]:
```
# Calculate average visits for each year and state
yearly_avg_241UDS0111 = df_241UDS0111.groupby(
    ["year_241UDS0111", "state"]
)["visits"].mean()

yearly_avg_241UDS0111
```

Out[12]:
```
year_241UDS0111  state
2022             Delhi          49749.00
                 Karnataka      45664.75
                 Maharashtra    50713.75
                 Tamil Nadu     43177.00
                 Telangana      49748.00
2023             Delhi          60031.75
                 Karnataka      55655.25
                 Maharashtra    60945.00
                 Tamil Nadu     55655.25
                 Telangana      60348.50
2024             Delhi          71202.50
                 Karnataka      68901.25
                 Maharashtra    73311.75
                 Tamil Nadu     66190.25
                 Telangana      72575.50
2025             Delhi          77465.50
                 Karnataka      78418.00
                 Maharashtra    83406.50
                 Tamil Nadu     75484.75
                 Telangana      83553.50
Name: visits, dtype: float64
```

In [ ]:
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## Step 14 – Using unstack()
```

In [13]:
```
# Reshape the grouped result: rows are years and columns are states
yearly_avg_241UDS0111 = df_241UDS0111.groupby(
    ["year_241UDS0111", "state"]
)["visits"].mean().unstack()

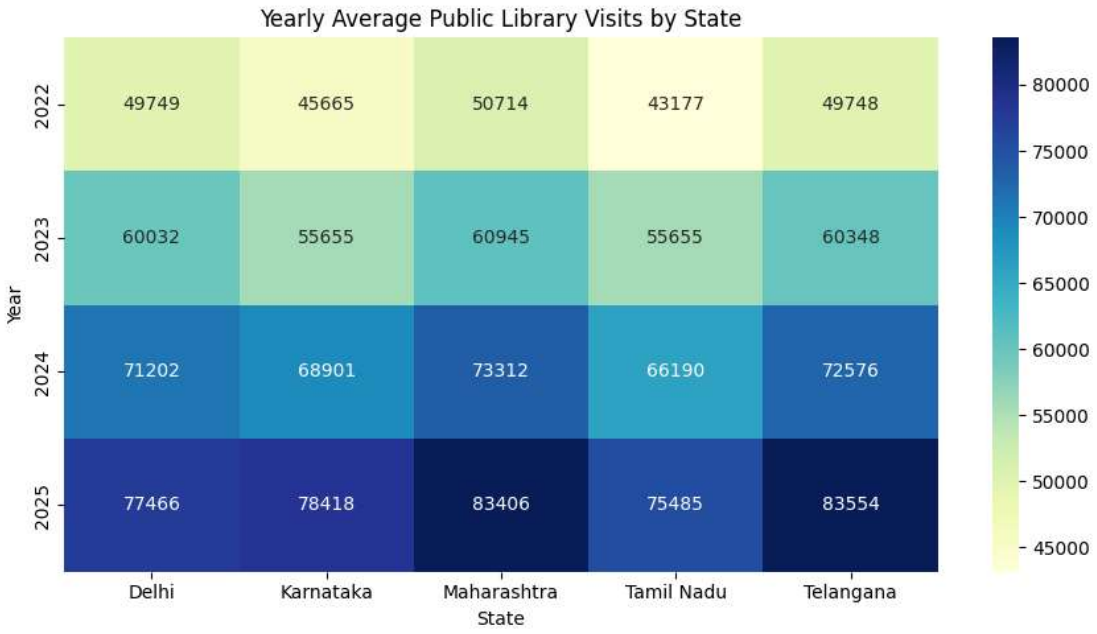
yearly_avg_241UDS0111.head()
```

Out[13]:

state	Delhi	Karnataka	Maharashtra	Tamil Nadu	Telangana
year_241UDS0111					
2022	49749.00	45664.75	50713.75	43177.00	49748.0
2023	60031.75	55655.25	60945.00	55655.25	60348.5
2024	71202.50	68901.25	73311.75	66190.25	72575.5
2025	77465.50	78418.00	83406.50	75484.75	83553.5

In [ ]:
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## Step 15 – Heatmap
```

In [14]:
```
# Heatmap: yearly average library visits by state
plt.figure(figsize=(9, 5), dpi=100)
sns.heatmap(
    yearly_avg_241UDS0111,
    fmt=".0f",
    annot=True,
    cmap="YlGnBu"
)
plt.title("Yearly Average Public Library Visits by State")
plt.xlabel("State")
plt.ylabel("Year")
plt.tight_layout()
plt.show()
```



In [ ]:
```
## Result
Various data visualization techniques using the Seaborn library were implemented to analyze and compare public
library visits across states and years.
```